

CASE REPORT

Toxicology; Pathology/Biology

Suicidal insulin injection combined with homicidal wrist incision

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Abstract

Determining fatal insulin overdose remains a major challenge in forensic practice due to the absence of characteristic histopathological findings and technical limitations in postmortem insulin detection. This report presented a complex case of a 69-year-old diabetic female involved in a suicide pact with her son, who was subjected to estazolam ingestion, excessive insulin injection, and a sharp-force incised wound of the left wrist with transection of the radial artery. She survived for more than 48 h after drug administration before ultimately succumbing to fatal sharp-force trauma inflicted by her son. Immunohistochemical staining confirmed insulin deposition at injection sites, and radioimmunoassay revealed markedly elevated insulin levels accompanied by a suppressed C-peptide concentration—findings consistent with exogenous insulin administration. Liquid chromatography–tandem mass spectrometry (LC–MS/MS) detected estazolam at a concentration of 18.2 ng/mL in blood. Forensic evidence confirmed the son's direct involvement at the scene. Based on the totality of the evidence, it was definitively concluded that the injuries were homicidal. This study offered valuable insights for the forensic identification and investigation of complex insulin-related fatalities.

KEYWORDS

estazolam, forensic pathology, forensic toxicology, insulin, poisoning, sharp-force injury

Highlights

- We report a complex case involving insulin overdose, estazolam ingestion, and sharp-force wrist injuries.
- Analysis of the competitive drug-injury relationship is crucial for determining the cause and manner of death.
- This complex case provides a key reference for the forensic investigation of insulin-related fatalities.

1 | INTRODUCTION

Insulin, a critical hormone in the regulation of glucose metabolism, plays an indispensable role in the management of diabetes [1]. However, its potent hypoglycemic effects can lead to severe

hypoglycemia and potentially fatal outcomes if used excessively [2–4]. In recent years, there has been a global rise in reported cases of suicide, homicide, and accidental deaths involving insulin. The forensic identification of insulin overdose poses significant challenges due to concealed injection sites, the absence of specific pathological



FIGURE 1 The scene findings. (A) A sharp instrument wound on the left wrist, characterized by disruption of blood vessels, skin shrinkage, redness, swelling, congestion, and blood clot formation at the wound edge. (B) An empty injection pen for “Insulin Aspartate 30”, and several empty packaging boxes of “Estazolam” were found at the scene.

changes, rapid postmortem degradation, difficulties in proper preservation, and the explanation of post-mortem insulin concentration [5–8]. Fatalities related to insulin often present complex characteristics resulting from multifactorial interactions, complicating the assessment of causality among competing mechanisms of death.

Excessive insulin primarily affects the central nervous system by inducing severe hypoglycemia, which can lead to cerebral edema, astrocyte proliferation, and ischemic alterations in neurons [9, 10]. The insulin-to-C-peptide ratio serves as a crucial indicator for assessing exogenous insulin administration [11]. An increased ratio can provide compelling evidence of exogenous insulin intake. In addition, immunohistochemistry has been utilized to detect abnormal insulin deposition at injection sites, thereby providing crucial evidence for evaluating exogenous insulin administration.

In this study, we present a case of insulin-related mortality characterized by substantial insulin injection, oral administration of

estazolam, and an incised wound of the left wrist with transection of the radial artery caused by a sharp instrument, which is involved in the competition for causes of death among the three factors. In forensic investigations, it is imperative to analyze this competitive relationship to accurately ascertain the cause of death and the nature of the case.

2 | CASE REPORT

2.1 | Case history

Upon receiving an emergency call and arriving at the scene, authorities discovered an elderly woman deceased in a hotel room, while a middle-aged man was stabilized and out of immediate danger after being rescued. At the scene, a blood-stained

utility knife, an empty injection pen for NovoMix 30, and several empty packaging boxes of “estazolam” were found (Figure 1B). The preliminary investigation revealed that the deceased was a 69-year-old diabetic patient who routinely administered 10IU of insulin NovoMix 30 twice daily. She and her son made an appointment to commit suicide together. She initially ingested 30 tablets of estazolam and concurrently administered a substantial dose of insulin. Her son ingested 40 tablets of estazolam. Following ingestion, both individuals remained in bed. During this period, the mother experienced frequent urination. Approximately 15 h later, they awoke and consumed a small amount of bread. The mother exhibited significantly altered mental status, characterized by slurred speech and an inability to ambulate, necessitating that her son carry her to the toilet. The son also experienced dizziness. After an additional 20 h, with both still alive, they discussed and mutually agreed to acquire a knife for the purpose of committing suicide. The son subsequently used a utility knife to incise his mother's wrist, resulting in substantial hemorrhage (Figure 1A). He then repeatedly lacerated his own wrist and stabbed himself in the neck. Following these actions, he slept for approximately 13 h. Upon regaining consciousness, he discovered that his mother had died.

2.2 | Autopsy and histopathological findings

Livor mortis presented as a light red discoloration on the uncompressible area of the back, while pallor was noted in the lips and face. An 8.0 cm transverse wound was observed on the anterior side of the left wrist. A flap was seen 0.5 cm near the ulnar incision angle at the proximal edge of the wound, and a 0.5 cm scratch was seen inward from the ulnar incision angle (Figure 1A). The incisions on both sides were characterized by sharp edges, with the wound cavity and walls appearing smooth and even, and no evidence of epidermal detachment at the wound margins. The wound cavity exhibited ruptures of the radial artery and multiple tendon lacerations. Additionally, numerous old pinhole-like injuries were visible on the abdomen, with two fresh pinholes located on the left and right sides of the umbilicus.

The left wrist suffered a sharp injury, severing blood vessels and causing skin shrinkage, redness, swelling, congestion, and clots at the wound's edge. Subcutaneous areas near fresh pinholes on both sides of the abdomen showed patchy hemorrhages of 1.0 cm × 1.0 cm and 0.8 cm × 0.6 cm. Microscopically, the dermis displayed edema, fused collagen, multiple hemorrhages, inflammatory cell infiltration, thrombi in vessels, cell clustering and forming protein-like substances (Figure 2C,D).

The brain appeared pale with segmental atherosclerotic plaques in basilar arteries, widespread edema, and ischemic changes in some neurons, including nuclear shrinkage, loss of Nissl bodies, increased eosinophils, and mineral particle deposition on the walls of some blood vessels around the hippocampus (Figure 2A,B). The heart

showed grade II-III atherosclerotic plaques in the left anterior descending and right coronary arteries. Significant necrotic atherosclerotic material was found in the plaques. The liver was earthy yellow with hepatocellular edema, anemic sinusoids, fibrous tissue hyperplasia, and lymphocyte infiltration in the portal area. The lungs showed edema with pale pink fluid in some alveoli, empty anemic capillaries, and occasional thrombi in small vessels. The spleen was atrophied and anemic, with anemic splenic sinuses and occasional thrombi. The pancreas was earthy yellow and anemic, with focal fibrosis, islet atrophy, and empty small vessels. The kidneys were pale and anemic, with some glomerular fibrosis, edematous renal tubular epithelial cells, and occasional thrombosis in the afferent arteriole and capillaries.

2.3 | Forensic evidence

Fingerprint analysis on the knife handle identified only the son's fingerprints. Furthermore, bloodstain analysis confirmed the presence of a mixed blood profile on the knife and surrounding surfaces, identifying both the mother's and the son's blood.

2.4 | Immunohistochemistry

Immunohistochemical (IHC) staining for insulin was conducted on skin samples from the insulin injection sites. The results revealed focal and patchy insulin-positive staining in the subcutaneous sweat glands, blood vessels, and adipose tissues on skin from the insulin injection sites (Figure 2E,F), while skin samples distant from the injection sites were negative.

2.5 | Detection of insulin and C-peptide

The concentrations of insulin and C-peptide were quantified using radioimmunoassay in samples of blood, liver tissue, pericardial effusion, subepithelial fat of the umbilicus, and periumbilical skin from both left and right sides. The homogenized tissues or serum samples and antibody are first incubated together to facilitate binding. Subsequently, radiolabeled antigen is added, which competes with the sample antigen for antibody binding sites, displacing some of the bound sample antigen. A secondary antibody that recognizes the primary antibody is then added, along with serum from the same species as the primary antibody, to promote flocculation and enable the separation of the antibody complexes from the solution by centrifugation. The radioactivity of the pellet is then measured to quantify the amount of radiolabeled antigen bound to the antibody, which allows for the determination of the original antigen concentration in the sample. The results demonstrated a significant elevation of insulin levels in the skin tissues at the suspected injection sites (Table 1).

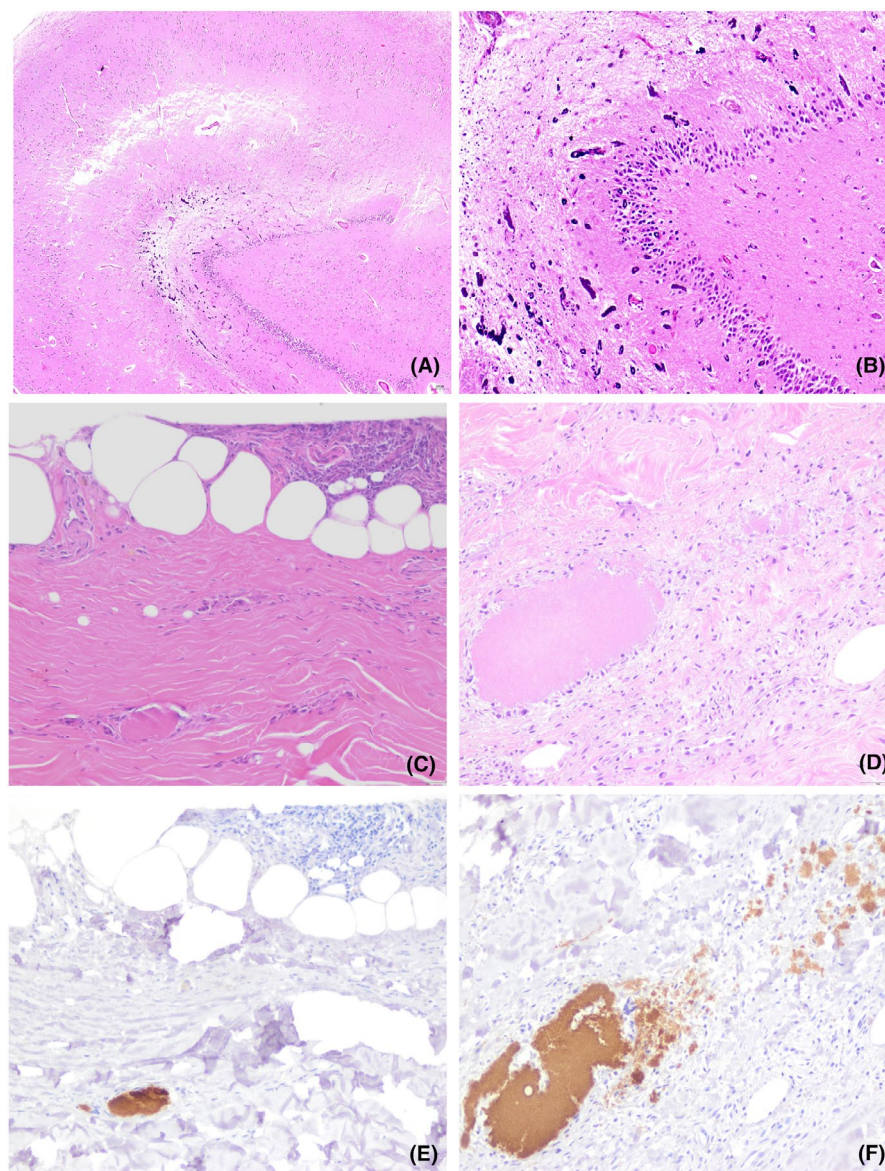


FIGURE 2 Histomorphological and immunohistochemical analysis. (A) Encephalomalacia and calcium salt deposition observed (HE, 40 $\times$). (B) Degeneration, necrosis, and calcium salt deposition in hippocampal neurons (HE, 200 $\times$). (C) Necrosis of collagen fibers with protein-like material in the right subcutaneous tissue of the umbilical cord (HE, 200 $\times$). (D) Aggregation of inflammatory cells with protein-like material in the right subcutis of the umbilicus (HE, 200 $\times$). (E, F) The left and right umbilical skin exhibited plaque-like positivity for insulin (IHC, 200 $\times$).

2.6 | Toxicological analysis

Estazolam was detected in postmortem blood samples using the LC-MS/MS method.

2.6.1 | Sample preprocessing

Serum (1 mL) was aliquoted, followed by the addition of 10 μ L of diazepam- d_5 and SKF-525A internal standard solution (100 ng/mL) and 2 mL of pH 9.2 borate buffer. The mixture was extracted with 4 mL of ether under vigorous mixing, then centrifuged. The supernatant was transferred and evaporated in a 60°C water bath

under a gentle stream of nitrogen. The residue was reconstituted in 100 μ L of mobile phase, and 5 μ L was injected into the LC-MS/MS system.

2.6.2 | LC conditions

Chromatographic separation was carried out on an LC-MS/MS 8045 system using a C18 column. The mobile phase comprised (A) 20 mmol/L ammonium acetate with 0.1% formic acid and (B) methanol. The column temperature was maintained at 25°C, with a flow rate of 0.35 mL/min. A gradient elution program was applied as follows: starting at 10% B, increasing linearly to 95% B over 3 min,

TABLE 1 Results of postmortem toxicology analysis.

Case	Glucose concentration(mmol/L) ^a	Insulin concentration(μU/mL) ^b	C-peptide(ng/mL) ^c	Estazolam(ng/mL)
Blood	18.02	<1.6	0.46	18.2
Liver	1.13	100.9	0.04	/
Pericardial effusion	6.57	<1.6	<0.03	/
The skin on the left side of the umbilicus	0.67	>300	<0.03	/
The skin on the right side of the umbilicus	0.25	>300	<0.03	/
Subepithelial fat of the umbilicus	<0.11	9.1	<0.03	/

Abbreviation: /, Not detected.

^aReference values: 3.90–6.10 mmol/L.

^bReference values: 1.8–11.8 μIU/mL.

^cReference values: 0.78–5.19 ng/mL.

maintaining for 3 min, rapidly returning to 10% B in 0.1 min, and followed by a 5.9-min re-equilibration period. The total run time was 12 min per injection.

2.6.3 | MS/MS conditions

Mass spectrometric analysis was conducted utilizing an electrospray ionization (ESI) source operated in positive ion mode with multiple reaction monitoring (MRM). Key instrument parameters were optimized as follows: dwell time, 10 ms; heating gas (air) flow, 10 L/min; nebulizing gas (nitrogen) flow, 3 L/min; drying gas (nitrogen) flow, 10 L/min; interface temperature, 300°C; interface voltage, 4 kV; desolvation line (DL) temperature, 250°C; and heat block temperature, 400°C.

2.6.4 | LC-MS/MS results

Estazolam was detected in the blood samples, with a concentration of 18.2 ng/mL. No other common toxic substances or drugs were detected (Table 1).

2.7 | Cause and manner of death

Based on a comprehensive analysis that included case investigation, autopsy findings, and toxicological analysis, it was determined that the cause of death was acute hemorrhagic shock due to the rupture of the radial artery from wrist cutting, following the administration of a substantial amount of insulin and the ingestion of estazolam. The manner of death was classified as homicide.

3 | DISCUSSION

Since the successful extraction of insulin in 1921, incidents of suicide, homicide, and accidental death resulting from insulin misuse

have been reported. The identification of insulin-related fatalities remains challenging in forensic investigations due to the covert nature of the crime and the absence of characteristic postmortem findings [12]. In forensic practice, these cases often involve multiple contributing factors, complicating the assessment of each factor's role in the death [13]. While previous studies have primarily focused on single-mechanism fatalities or epidemiological reviews of insulin overdose, cases involving combined mechanisms—particularly the concurrence of insulin toxicity with drug ingestion and mechanical injury—remain poorly documented. This report presents a 69-year-old female who succumbed to a combination of excessive insulin injection, estazolam ingestion, and incised wound of the left wrist with transection of the radial artery. A critical objective of this investigation is to clarify the interplay among these factors and to determine the manner of death.

Multiple pinpoint-like marks were observed on the abdominal wall, along with elevated insulin levels and subcutaneous hemorrhage at the injection sites. An empty NovoMix 30 pen was found at the scene, indicating significant insulin administration before death. However, several factors contradict fatal insulin overdose as the direct cause. First, the victim survived more than 48 h after injection, exceeding the typical 22-h duration of action for NovoMix 30 [14]. Second, histopathology of injection sites showed local tissue reactions (inflammatory infiltration, collagen necrosis, protein exudation), but no evidence of systemic fatal effects. Third, no widespread ischemic neuronal necrosis was found in the hippocampus. Additionally, cardiac blood contained estazolam at 18.2 ng/mL—within the therapeutic range (5–50 ng/mL) [15]. Given the drug's pharmacokinetics and the over 48-h survival period, estazolam alone is unlikely to have caused death. Postmortem findings included multi-organ pallor, cerebral edema, and ischemic neuronal changes. A sharp-force injury on the left wrist exhibited signs of vital reaction, including tissue contraction, erythema, and clot formation. A bloodstained knife was found at the scene. Based on the integrated analysis, the immediate cause of death was acute hemorrhagic shock due to transection of the radial artery, which occurred following the ingestion of estazolam and insulin.

Insulin's short half-life (approximately 4–6 min) and susceptibility to postmortem degradation, hemolysis, and putrefaction complicate its detection and quantification. To date, there is no universally accepted method for postmortem insulin analysis [16, 17]. The principal techniques developed for insulin detection include immunoassays, chromatographic methods, and electrochemical biosensors [18–20]. Among these, LC–MS/MS provides high specificity and selectivity, enables the identification of metabolites, and demonstrates strong resistance to interference; however, it involves complex and costly sample preparation procedures [21]. The limitations of immunoassays include susceptibility to interference from anti-insulin antibodies, the absence of an international standardized methodology, and the inability to differentiate between specific insulin analogs. Insulin was measured by RIA (upper reportable limit 300 μ U/mL). For samples above range, serial dilutions (up to 1:100) were performed to bring results within calibration; back-calculated values showed acceptable agreement ($\approx$ 90–110%). Routine high/low controls were within limits; a full method validation (LOD/LOQ and comprehensive precision studies) was beyond the scope of this casework. This approach confirmed markedly elevated insulin levels at injection sites alongside low C-peptide—consistent with exogenous administration. However, due to the narrow analytical range of the assay, precise quantification was not achievable.

A review of 66 insulin overdose cases indicated that hypoglycemia (blood glucose <2.6 mmol/L) is a key indicator of insulin-induced fatality [6]. However, postmortem blood glucose levels are unreliable due to glycogenolysis and gluconeogenesis, limiting their diagnostic value. In this case, postmortem blood glucose measured 48 h after death was 18.02 mmol/L—far exceeding normal antemortem levels—highlighting the limited utility of postmortem glucose analysis. Antemortem glucose measurements are far more informative in such contexts. To better evaluate possible antemortem hypoglycemia, alternative biomarkers should be considered. β -hydroxybutyrate levels, for instance, may help indicate antemortem hypoglycemia, as elevated levels can reflect metabolic stress during insulin-induced hypoglycemia [22]. In comparable cases, elevated β -hydroxybutyrate may reflect metabolic stress during insulin-induced hypoglycemia, and HbA1c can contextualize longer-term glycemic control. These biomarkers were not obtained in the present case; therefore, our references to them are conceptual and not derived from case-specific measurements.

This case presents a rare and instructive example of a staged suicide involving multiple self-harm mechanisms. The combination of insulin overdose, estazolam ingestion, and a subsequent sharp-force injury is exceptionally uncommon in the forensic literature. Furthermore, this case is distinguished by a prolonged survival interval of over 48 h, which contrasts sharply with typical acute insulin fatalities that usually prove fatal within hours. This extended timeline indicates that the initial actions were insufficient to cause rapid death, ultimately leading to the final, fatal injury. This dynamic not only highlights the unique features of the present case but also

underscores the critical importance of reconstructing the sequence of events in complex forensic investigations.

Based on scene investigation and forensic analysis, it was concluded that the victim and her son had made an appointment to commit suicide together. The victim first ingested estazolam and self-administered a large insulin dose. The combined sedative and hypoglycemic effects induced profound central nervous system depression, rendering her weak and powerless but not causing immediate death. She remained in a poor mental state for approximately 48 h before her son inflicted the fatal incised wound of the left wrist with transection of the radial artery. Stephenson et al. reported that 82.5% of insulin overdose fatalities were attributed solely to insulin toxicity, while 17.5% involved mixed drug toxicity—a pattern also observed in other cases, including two instances of fatal insulin homicide where psychotropic drugs were detected [23].

Determining the cause and manner of death in multifactorial cases involving insulin intoxication presents significant challenges for forensic experts and judicial systems. To address these complexities, the following considerations are essential: (1) a thorough scene investigation to uncover often-concealed evidence such as syringes, vials, or injection devices, coupled with a meticulous autopsy examination for injection sites; (2) the proper handling of biological specimens, which must be collected and frozen immediately to ensure analyte stability; (3) a tissue specimen of approximately 50 g should be excised from the soft tissue surrounding the puncture site, covering a defined 5 $\times$ 5 cm area centered on the puncture. For comparative analysis, a control specimen of equal quantity should be harvested from the symmetrical location on the contralateral side of the body; (4) the comprehensive application of analytical techniques, including LC–MS/MS and electrochemical biosensors, to accurately detect insulin and its metabolites; and (5) an integrated assessment that synthesizes toxicological, pathological, and investigative findings to disentangle the contributions of multiple potential causes of death.

4 | CONCLUSION

This report details a complex case of suicide involving insulin overdose, estazolam ingestion, and an incised wound of the left wrist with transection of the radial artery. The prolonged survival period distinguishes it from typical acute insulin fatalities and highlights the necessity of reconstructing the timeline of events. Our findings affirm that an integrated forensic approach—correlating scene evidence, autopsy, toxicology, and pathology—is paramount for accurate determination in multi-factorial deaths. Future work should focus on standardizing postmortem insulin analysis and collecting similar cases to refine interpretive protocols.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

REFERENCES

- Vanderpant N, Ward E, Farrell E, Theodoraki A. Insulin for people with type 2 diabetes mellitus. *BMJ*. 2024;386:e78015. <https://doi.org/10.1136/bmj-2023-078015>
- Hoeg-Jensen T, Kruse T, Brand CL, Sturis J, Fledelius C, Nielsen PK, et al. Glucose-sensitive insulin with attenuation of hypoglycaemia. *Nature*. 2024;634(8035):944–51. <https://doi.org/10.1038/s41586-024-08042-3>
- Gouveri E, Gkouveri A, Popovic DS, Papazoglou D, Papanas N. Intentional insulin overdose and depression in subjects with and without diabetes mellitus: a commentary. *Diabetes Ther*. 2024;15(9):1845–54. <https://doi.org/10.1007/s13300-024-01623-5>
- Endall R, McCallum R, Burgess J. Case report: high-concentration insulin glargine overdose complicated by hepatic steatosis. *J Endocr Soc*. 2020;4(5):bvz020. <https://doi.org/10.1210/jendso/bvz020>
- Yuan Y, Yu Z, Tong F, Zhao S, Li Y, Shi Q, et al. A retrospective study of 29 fatal cases of insulin overdose. *Forensic Sci Int*. 2024;361:112126. <https://doi.org/10.1016/j.forsciint.2024.112126>
- Marks V. Murder by insulin: suspected, purported and proven—a review. *Drug Test Anal*. 2009;1(4):162–76. <https://doi.org/10.1002/dta.38>
- Yu Z, Yuan Y, Zhang J, Li Y, Wang Z, Wang Y, et al. Review of the lethal mechanism of insulin poisoning and the characteristic of forensic identification. *Leg Med (Tokyo)*. 2024;70:102478. <https://doi.org/10.1016/j.legalmed.2024.102478>
- Arbouche N, Farrugia A, Gheddar L, Ameline A, Blanchot A, Raul JS, et al. Is it really possible to kill with insulin without leaving traces? From lifesaver to killer, the issues surrounding the analytical characterization of postmortem insulin illustrated by an exemplary case. *J Forensic Sci*. 2024;69(3):1106–13. <https://doi.org/10.1111/1556-4029.15501>
- Suh SW, Hamby AM, Swanson RA. Hypoglycemia, brain energetics, and hypoglycemic neuronal death. *Glia*. 2007;55(12):1280–6. <https://doi.org/10.1002/glia.20440>
- Hess C, Musshoff F, Madea B. Disorders of glucose metabolism—post mortem analyses in forensic cases: part I. *Int J Leg Med*. 2011;125(2):163–70. <https://doi.org/10.1007/s00414-010-0509-6>
- Iwase H, Kobayashi M, Nakajima M, Takatoril T. The ratio of insulin to C-peptide can be used to make a forensic diagnosis of exogenous insulin overdosage. *Forensic Sci Int*. 2001;115(1–2):123–7. [https://doi.org/10.1016/s0379-0738\(00\)00298-x](https://doi.org/10.1016/s0379-0738(00)00298-x)
- Tong F, Wu R, Huang W, Yang Y, Zhang L, Zhang B, et al. Forensic aspects of homicides by insulin overdose. *Forensic Sci Int*. 2017;278:9–15. <https://doi.org/10.1016/j.forsciint.2017.06.015>
- Handlosova K, Sistik P, Uvira M, Andelova K, Handlos P, Stejskal D. Contribution of methamphetamine and insulin to the death of a woman suffering from type I diabetes – which played the greater role? *Forensic Sci Med Pathol*. 2025;21(1):502–6. <https://doi.org/10.1007/s12024-024-00855-y>
- Christiansen JS, Niskanen L, Rasmussen S, Johansen T, Fulcher G. Lower rates of hypoglycemia during maintenance treatment with insulin degludec/insulin aspart versus biphasic insulin aspart 30: a combined analysis of two phase 3a studies in type 2 diabetes. *J Diabetes*. 2016;8(5):720–8. <https://doi.org/10.1111/1753-0407.12355>
- Drummer OH. Postmortem toxicology of drugs of abuse. *Forensic Sci Int*. 2004;142(2–3):101–13. <https://doi.org/10.1016/j.forsciint.2004.02.013>
- Heald AH, Bhattacharya B, Cooper H, Ullah A, McCulloch A, Smellie S, et al. Most commercial insulin assays fail to detect recombinant insulin analogues. *Ann Clin Biochem*. 2006;43:306–8. <https://doi.org/10.1258/000456306777695690>
- Manley SE, Stratton IM, Clark PM, Luzio SD. Comparison of 11 human insulin assays: implications for clinical investigation and research. *Clin Chem*. 2007;53(5):922–32. <https://doi.org/10.1373/clinchem.2006.077784>
- Sunderland N, Wong S, Lee CK. Fatal insulin overdoses: case report and update on testing methodology. *J Forensic Sci*. 2016;61(1):S281–S284. <https://doi.org/10.1111/1556-4029.12958>
- Vargas E, Nandhakumar P, Ding S, Saha T, Wang J. Insulin detection in diabetes mellitus: challenges and new prospects. *Nat Rev Endocrinol*. 2023;19(8):487–95. <https://doi.org/10.1038/s41574-023-00842-3>
- Soffe R, Nock V, Chase JG. Towards point-of-care insulin detection. *ACS Sens*. 2019;4(1):3–19. <https://doi.org/10.1021/acssensors.8b01253>
- Shen Y, Prinyawiwatkul W, Xu Z. Insulin: a review of analytical methods. *Analyst*. 2019;144(14):4139–48. <https://doi.org/10.1039/c9an00112c>
- Li K, Wang W, Wu J, Xiao W. β -Hydroxybutyrate: a crucial therapeutic target for diverse liver diseases. *Biomed Pharmacother*. 2023;165:115191. <https://doi.org/10.1016/j.biopha.2023.115191>
- Stephenson L, Van Den Heuvel C, Humphries M, Byard RW. Characteristics of fatal insulin overdoses. *Forensic Sci Med Pathol*. 2022;18:429–41. <https://doi.org/10.1007/s12024-022-00511-3>

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